

K-Means Clustering: From First Principles

Why Do We Need Clustering?

Many datasets do not come with labels. A store may have customer purchase history but no labels saying which customer type each person belongs to. A company may have website behavior data but no categories. In these cases, supervised learning cannot be used directly because there is no target column.

Clustering helps us discover natural groups in data. K-Means is one of the simplest and most important clustering algorithms.

Supervised vs Unsupervised Learning

In supervised learning, the model learns from inputs and known outputs. Linear Regression learns from house features and house prices. Logistic Regression learns from email features and spam labels.

In unsupervised learning, there are no target labels. The algorithm only sees the input data and tries to find structure.

K-Means is unsupervised because it groups data without being told the correct groups.

What Is K-Means?

K-Means is a clustering algorithm that divides data into K groups. Each group is called a cluster.

The algorithm tries to put similar points in the same cluster and different points in different clusters.

The center of each cluster is called a centroid. A centroid is not always an actual data point. It is the average position of the points assigned to that cluster.

The Core Idea

K-Means repeats two steps: assignment and update.

In the assignment step, each data point is assigned to the nearest centroid.

In the update step, each centroid moves to the average position of the points assigned to it.

The algorithm repeats these steps until the centroids stop moving much or the assignments stop changing.

Distance Calculation

K-Means usually uses Euclidean distance.

For two points in two dimensions, the formula is:

$$Distance = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

The nearest centroid is the centroid with the smallest distance.

Because distance is so important, feature scaling matters. If one feature has much larger values than another, it can dominate the clustering result.

Assignment Step

Suppose there are three centroids. For each data point, K-Means calculates the distance to each centroid and assigns the point to the closest one.

If a customer is closest to centroid 2, the customer is assigned to cluster 2.

The assignment step creates groups based on the current centroid positions.

Update Step

After assignment, K-Means recalculates each centroid.

The new centroid is the average of all points in that cluster.

For one-dimensional data, the centroid is:

$$Centroid = \frac{x_1 + x_2 + \dots + x_n}{n}$$

For multi-dimensional data, the average is calculated separately for each feature.

This moves the centroid toward the center of its assigned points.

Convergence

K-Means keeps repeating assignment and update. Eventually, the centroids stop changing much. At that point, the algorithm has converged.

Convergence does not always mean the clustering is perfect. It means the algorithm has reached a stable solution based on its starting centroids.

Because starting positions matter, K-Means is often run multiple times with different initial centroids.

Inertia

Inertia measures how tightly points fit inside their clusters. It is the sum of squared distances between each point and its assigned centroid.

$$Inertia = \sum_{i=1}^n \|x_i - c_i\|^2$$

Here, x_i is a data point and c_i is the centroid of the cluster assigned to that point.

Lower inertia means points are closer to their centroids. But inertia always decreases as K increases, so it cannot be used alone without judgment.

Choosing K and the Elbow Method

K is the number of clusters. Choosing K is one of the most important decisions in K-Means.

If K is too small, different groups may be forced together. If K is too large, one natural group may be split into unnecessary smaller groups.

The Elbow Method compares inertia for different values of K. At first, increasing K reduces inertia a lot. Later, improvement becomes smaller. The elbow point is where the improvement starts slowing down.

The elbow suggests a reasonable K, but it is not always obvious. Domain understanding still matters.

Worked Example

Suppose a store wants to group customers using annual income and spending score. K-Means may discover groups such as low income with low spending, high income with high spending, and high income with low spending.

The algorithm does not know these names. It only finds clusters based on distance. Humans interpret the clusters afterward.

This is important: K-Means finds mathematical groups. We give business meaning to those groups.

Advantages, Disadvantages, and Evaluation

K-Means is simple, fast, and easy to understand. It works well when clusters are roughly round and similarly sized.

Its disadvantages are that K must be chosen, it is sensitive to scaling, sensitive to outliers, and can struggle with non-spherical clusters.

K-Means is evaluated using clustering metrics such as inertia and Silhouette Coefficient. Since there are no true labels in unsupervised learning, evaluation is more indirect than supervised learning.

Use K-Means when you want a simple clustering method and distance-based grouping makes sense.
Avoid it when clusters have complex shapes or many outliers.